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11 August 1965
SUPERSEDING
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13 January 1961

MILITARY SPECIFICATION

PROPELLANT, M5, FOR USE IN AND CARTRIDGE, 90MM, HEAT, M371 & M371E1

1. SCOPE

1.1 This specification covers the M5 propellant for use in the 90MM HEAT Cartridge, M371, M371E1 and 90MM Practice M371.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-652 - Propellants, Solid For Cannons Requirements and Packing

FSC: 1375

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWING

ARMY

8848808 - Charge, Propelling, XM82
8863468 - Cartridge, 90MM, HEAT, M371E1 w/Fuze for 90MM Rifle, M67 Loading, Assembling and Marking Diagram

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and Revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite 49 CFR 71-90 (latest revision).

3. REQUIREMENTS

3.1 Material.-The propellant shall be M5, Type II and conforming to the requirements of Standard MIL-STD-652.

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3.2 Ballistics.-The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured shall comply with the following requirements when tested as specified in 4.3.1.

3.2.1 Velocity and pressure.-At 70 degrees Fahrenheit (F.) a weight of propellant shall be charge assessed to comply with the prescribed service velocity of 700 feet per second (213.4 meters per second) without exceeding a maximum rated pressure of 7,800 pounds per square inch (psi), piezo equivalent.

3.2.2 Uniformity of velocity.-The standard deviation in velocity for the lot at 70 degrees F. shall not exceed 9.0 feet per second (2.74 meters per second).

3.2.3 Uniformity of maximum pressure.-The average maximum pressure produced plus four (4) times the standard deviation of the maximum pressure obtained in the uniformity series shall not be greater than 115 percent of the rated maximum pressure of the gun (8,970 psi, piezo equivalent).

3.3 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

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4.2.1 Lot formation.-A lot shall consist of one or more batches of M5 propellant produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M5 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. Product shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235 when applicable.)

4.2.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect.

4.2.3 Testing

4.2.3.1 Ballistics (see 3.2).-A sample of 50 pounds of the propellant shall be randomly selected from the lot for charge establishment, determining uniformity or velocity and determining uniformity of maximum pressure.

4.2.3.1.1 Pressure (see 3.2.1) Critical defect, Code No. 01001.-If the mean of the maximum pressures of the test rounds, when calculated and corrected to the corresponding piezo equivalent at service velocity, exceeds the applicable requirement, the lot shall be rejected.

4.2.3.1.2 Uniformity of velocity (see 3.2.1 and 3.2.2) Major defect, Code No. 02001.-If the product of the factor 0.73 times the velocity standard deviation of the test rounds exceeds the applicable requirement, the lot shall be rejected (see 6.2).

4.2.3.1.3 Uniformity of maximum pressure (see 3.2.3) Critical defect, Code No. 03001.-The lot of propellant shall be rejected if it fails to meet the applicable requirement (see 6.2).

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4.2.3.2 Retest.-If for any reasons the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.3 Test methods and procedures

4.3.1 Ballistic tests.-These tests shall be conducted at a Government proving ground in accordance with the applicable Ordnance Proof Manual and/or Acceptance Test Procedure. A weight of charge shall be calculated, from charge velocity and charge-pressure curves representing actual firing data, to give prescribed service velocity. A ten round uniformity series shall be fired and the velocity mean and standard deviation shall be calculated after reducing results to standard conditions and pressure. The pressure shall be reported as piezo equivalent pressure.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-The propellant shall be packed and marked in accordance with MIL-STD-652.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1).
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation.-The standard deviation shall be calculated with (N-1) as the divisor in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, and one copy to ATTN: SMUPA-DC8, Dover, New Jersey.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number 1375-A945

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